Access network life-cycle costs

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The cost analysis associated with the successful introduction of any new technology into BT's network is crucially important. This is particularly true of the access network where each individual customer requires dedicated physical connectivity. This paper details the importance of access network cost analysis by considering a life-cycle model. The holistic approach of life-cycle costs is promoted, indicating some of the principles and methods that can be adopted.

The paper endeavours to give a flavour of some of the access network cost modelling that has been performed by BT indicating the benefit and relationship with regard to the design and implementation life cycle of the access network.

1. Introduction

The access network is the 'last link' between the telecommunications network and the customer. Although the 'last link' may only consist of a relatively small distance (usually less than 7.5 km in the UK), it represents a significant portion of the overall cost in order to provide and maintain a customer with network connectivity. Indeed, France and Mackenzie [1] suggest that the access network typically accounts for approximately 80% of all network investment. Young et al [2] confirm this huge proportion and suggests that BT's current access network has a replacement cost of £11 billion. The main reason for this cost is the fact that each individual customer requires dedicated physical connectivity, whereas elsewhere, particularly in the core network, bandwidth requirements are aggregated and as such the economics of large bit rate transmission systems using optical fibre may be applied.

Hence, the access network is very sensitive to factors affecting the cost. Indeed, the one common factor that any successful introduction of new access technology will have, is a thorough study of the costs attributable to them. This study will provide an integral part of any business or investment case [3]. It is crucial that any business case includes all aspects of the costs associated with the technology. Failure to do so will lead to at best reduced profitability or cost benefit, and at worst the overburdening costs of inappropriate technology, which in turn will lead to an inability to compete in an increasingly aggressive market.

Thus, a holistic approach to cost modelling is required. Bell et al [4] highlighted the important role of the telecommunications life cycle in providing this holistic view. The remainder of this paper will consider this life cycle in more detail, while also discussing the main principles, methodologies and techniques that are used in any accurate cost analysis. Finally, this paper endeavours to give a flavour of some of the cost analysis performed at some of the stages of the life cycle in order to ensure a cost-effective access network.

2. The telecommunications life cycle

Figure 1 illustrates the five main distinct stages of any telecommunications network life cycle [4]. Costs associated with each stage should be incorporated in any cost analysis in order to give a comprehensive understanding of the cost factors and provide opportunities for continued cost reduction. The significance and role of the cost analysis during these phases is now discussed.

2.1 Requirements capture

During the requirements capture phase, initial cost data is collated and analysed to aid the development of the statement of requirements (SoR). This cost analysis may, for example, indicate the expected cost per customer served, and the relationship between fixed and incremental costs.

Furthermore, preliminary cost analysis is performed and input into the business case for an investment appraisal. The investment appraisal is inextricably linked and critically dependent upon consistent and accurate forecasting of the costs. The cost analysis and investment appraisal represents a crucial factor in managing, at an initial stage, network investment. For example, BT spent £2.5 billion in 1991/92, on tangible fixed assets; it is essential to the company's

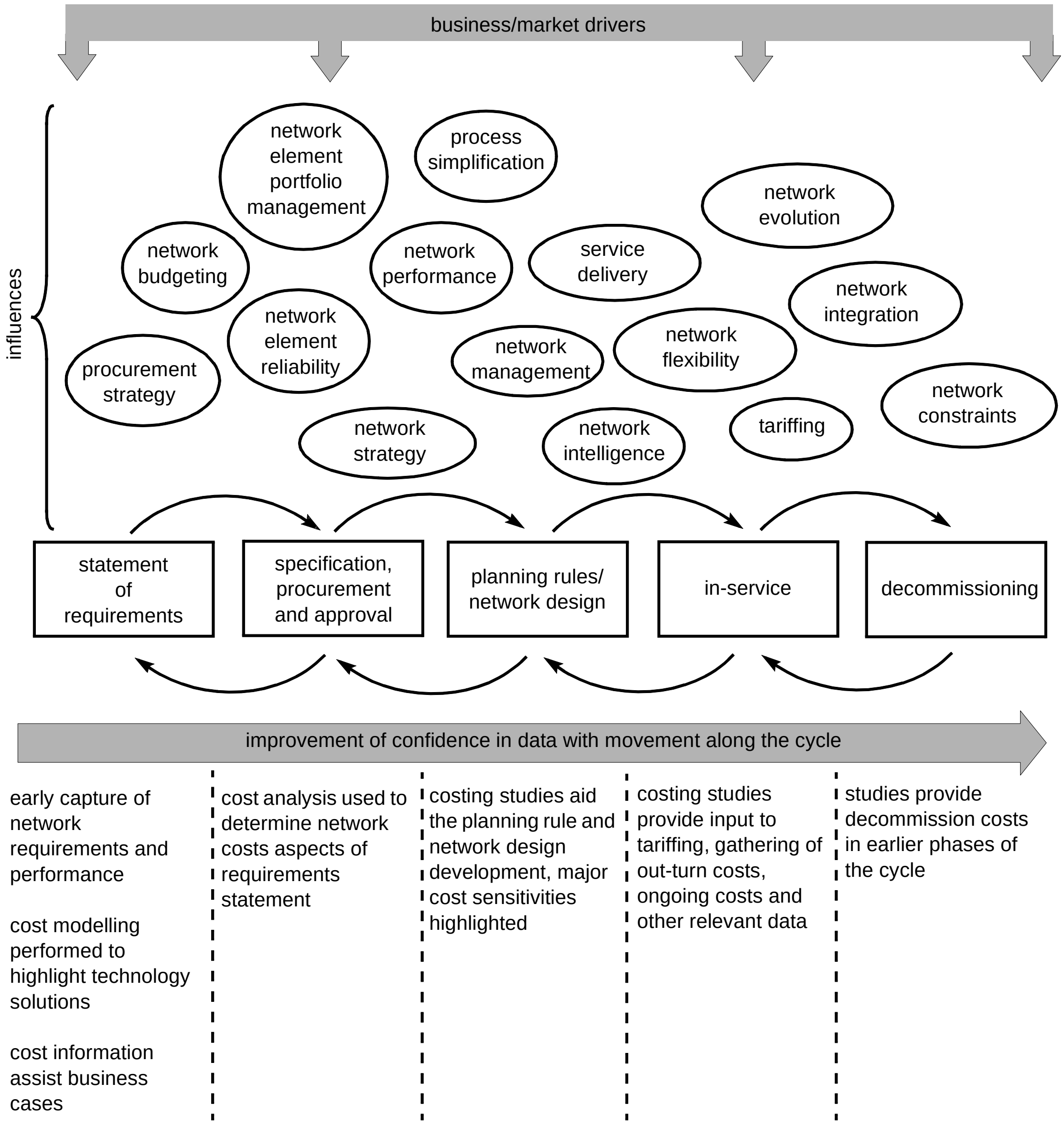


Fig 1 The telecommunications network life cycle.

success therefore that this investment is managed well [3]. Moreover, it is estimated that approximately 80% of the whole-life costs are committed during the requirements, specification and design stages of the life cycle [5, 6]. Therefore the earlier any potential non-profitable network expenditure is highlighted, the more opportunity there will be to redirect the funds to more lucrative investments.

It follows that an early analysis of the costs during the requirements stage of the life cycle provides a critical tool for early decision making and investment appraisal, while also indicating cost factors for the SoR.

2.2 *Specification*

In order to fully specify the network based upon the SoR, capital and current account aspects of the network costs (see section 3) must be fully understood. This provides a powerful tool in balancing the network specification against the costs, i.e. the network may be specified with a number of criteria indicated in the SoR. However, the extent to which each of these criteria is met, may have a profound effect upon the final cost. One example of this is the trade-off between the network reliability and initial deployment cost. Although the network may have to meet some minimum reliability as detailed in the SoR, it may be desirable from an overall cost perspective to have a network specified to a much higher level of reliability — or conversely, the reliability requirements may, upon further investigation, be so constraining that it incurs a heavy cost penalty.

Once the network has been specified, an invitation to tender (ITT) may be issued. As part of the selection process comprehensive cost investigations are performed. These cost investigations will not only include an overall cost comparison between the proposals but also indicate cost sensitivities, which, used in conjunction with an examination of how the proposals meet the specification, will lead to a competent selection strategy.

Finally, during the actual procurement process a cost analysis will indicate competitive pricing of network components, in order to confidently perform portfolio management.

2.3 Network design and planning

The planning stage of the life cycle encompasses planning and network design at multiple levels. Ward [7] refers to seven main types of planning:

- strategic planning — this provides an overall framework within which the planning of individual networks must conform (any strategic plan must take account of the influences highlighted in Fig 1),
- conversion planning — this represents a plan of how to evolve the network to the future network detailed in the strategic plan,
- network engineering planning — this provides the design of the network to support specific new services,
- enhancement planning — identifies opportunities to enhance the network to reduce cost, improve performance and increase its flexibility and response to growth and new service opportunities,
- long-term planning and programme management — these provide more detail of the requirements upon a specific exchange for example, in order to meet the vision indicated in the strategic plan,
- short-term planning — this comprises continuous review and re-planning of the system as conversion and enhancement plans are implemented to meet growth,
- implementation plans (sometimes referred to as work plans) — these provide further details of how to actually implement some of the plans previously discussed.

Analysis of the cost sensitivities and key drivers is required at all these different levels of planning. A successful network strategy and deployment, brought about by appropriate planning, will take account of key cost drivers to minimise the expense to the network provider. Failure to provide an adequate deployment strategy can undermine previous work and induce additional network expense, resulting in the original business objectives not being achieved.

2.4 In-service

After the network has been deployed and the desired services are running over it, the life cycle is in the in-service phase. The in-service costs are arguably less widely documented and studied than the costs associated with network deployment and design, although they can dominate and determine long-term profitability. As indicated earlier, this is particularly true of the access network where the scale of the in-service operations is so large.

A major focus of cost investigations at this stage is to aid what is known as process re-engineering. Process re-engineering is the practice of determining improved methods of performing tasks. For example, can improvements be made into how faults in the network are dealt with? Process re-engineering has come to the fore in recent years, due to the potential savings that can be gained in optimising the deployment, management and running of a telecommunications network. (It should be noted that process re-engineering is not just limited to the telecommunications industry and has been utilised in a vast number of other industries.) In order to perform such studies, the costs associated with the processes (usually current account costs, see section 3) are a key driver to optimisation, i.e. the main objective is generally to optimise the processes and thereby reduce the operating costs. Consequently a thorough understanding of the costs associated with the running and management of the network is of prime importance during the in-service phase (see section 4.3).

2.5 Decommissioning

The decommissioning phase, not considered by Bell et al’s discussion of access network costs [4], represents an integral part of the telecommunications life cycle. The costs associated with the decommissioning of a network must be incorporated in any cost analysis and investment appraisal, as they can have an impact upon any future proposals. For example, the duct used in an obsolete network could be reutilised by any proposed network provided the original cables are removed. Therefore, the costs associated with decommissioning can have a serious impact on the opportunities for future exploitation of deployed networks.

Furthermore, the truncation of unexpired depreciation of assets needs to be taken into account, since it may significantly impact upon the costs associated with the introduction of new technology.

The costs associated with decommissioning should therefore be incorporated into any investment appraisals performed during the statement of requirements stage.

3. Cost methodologies

The costs associated with each phase of the telecommunications life cycle can be split into either capital or current account costs. A precise definition of capital and current account costs is difficult and subject to interpretation due to the inconsistency in costing terminology and individual company accounting policies. However, for the sake of simplicity the following definitions, which broadly agree with Ward [8], shall be applied for the rest of this paper:

- capital account costs are those costs associated with the initial purchase and deployment of the network (or installed first costs (IFC) [9]),
- current account costs are those costs incurred in operating the network.

Current account costs may be further subdivided into non-recurring current account (NRCA) referring to one-off costs and recurring current account (RCA) referring, as the term suggests, to repetitive costs [3].

The practice of accurately determining the capital and current account costs represents a challenging discipline in any industry — however, the telecommunications industry suffers from the following effects which make the exercise of determining costs particularly problematic [10]:

- most products share the use of the same network or support structure, and costs do not vary significantly with volume usage,
- different products and services are subject to different levels of competition — there are few natural markets in which prices can be established and therefore prices tend to be cost-based and subject to the Regulator (this increases the importance of the role of cost analysis during the in-service phase),
- the different products consume the available resources in different proportions.

Fortunately, there are a number of techniques and methodologies which exist to aid the analysis of costs. These cost methodologies and techniques are generic to any costing exercise. This section highlights the main, but by no means all, the techniques and methodologies available.

3.1 ABC methods

A thorough analysis of costs may be performed using activity based costings (ABC) [10—13]. ABC is principally a method developed by accountants and promoted by the Harvard Business School, as a response to the challenge issued by Professor Kaplan:

> “.. devise new internal accounting systems that will be supportive of the firm’s new manufacturing strategy.”

ABC represents a methodology of accounting for costs based upon capturing all the operations and resources required in order to supply a service. Thus, ABC analysis includes taking into account the development, deployment and operating costs to maintain a particular service, i.e. ABC methods would incorporate costs expended during all phases of the life cycle. Bussey [10] suggests that ABC costs are similar to the method that BT employs of cost causation, i.e. the costs are allocated on the basis of what caused them to be incurred; for example, the installation of specific lines for specialised services will incur costs which are simple to identify.

In this way ABC provides a mechanism for accurately accounting for costs associated with providing a particular service. However, by its very nature it is complex and difficult to collate all the required information. Moreover, a number of nuances, specific to the access network, prevail to increase the complexity. For example, considering the introduction of digital subscriber line (DSL) technologies providing broadband service over copper which traditionally provided PSTN services only, the question of how the costs associated with the copper should be apportioned needs to be addressed. Clearly it would be unfair to allocate all the costs of the copper to the initial PSTN service. Furthermore, if the costs are allocated in some fashion, it may be performed in several ways — by bandwidth, by service or by technology (i.e. the DSL technology could carry several services). Also the allocation of costs adds a further complication — how is account taken of the costs associated with the practice of capacity planning, i.e. the costs associated with that portion of the cable which has been reserved for future service requirements?

Another drawback with ABC is that it is particularly geared towards analysing an existing cost structure. It is very difficult to utilise such an analysis to predict the costs associated with a new business process, as may be required for a new network technology or service. Indeed, as the difference between the new process and the existing one increases, so the usefulness of the ABC approach diminishes.

3.2 Whole-life costs

Whole-life cost techniques are closely aligned to the ABC methodology. Whole-life costs refers to those costs incurred during all the phases of the life cycle. While cost analysis of the individual phases may be considered in isolation, it is only by studying the costs associated with all the phases of the life cycle that a true and comprehensive comparison of different access network solutions and scenarios may be performed, and the associated successful management decisions taken.

Davies and Napier [5, 14] liken costs to an iceberg where the acquisition or deployment cost associated with any business venture represents the tip, and the larger, submerged, portion is the cost associated with ownership. The techniques of considering whole-life costs was originally applied by the US Department of Defense in the late 1960s, who realised that the cost of ownership were far greater than the initial purchase cost. Davies and Napier cite the example of the US defense budget of $126 billion, of which only 26% was procurement of new products (capital account), and 61% was operation and support (current account).

Thus, whole-life costs represent a true perspective of the costs incurred in any business options considered. They include costs associated with all the phases of the life cycle. However, as with the ABC methodologies, determining the whole-life cost of the access network is immensely difficult. Indeed, Davies and Napier [5, 14] suggest that there was little evidence of UK managers utilising whole-life methodologies as late as the 1980s. Davies and Napier went on to say that more recently a number of tools have been developed to consider life-time costs, e.g. the WINA tool used by BT, or the CASA tool used by the US Department of Defense. The appropriate use of these tools can be very powerful in providing cost analysis in investment appraisal activities.

3.3 End-to-end costs

An investigation of access network costs may be split into several areas, with each area considered separately. However, in order to compare different solutions and scenarios, the whole network cost must be considered. For example, the costs associated with terrestrial network solutions may be split broadly into two categories — electronics costs (e.g. line-cards in exchange equipment), and physical plant costs (e.g. cable, ducts and nodes). While these two categories may be studied separately, there are significant interrelationships, and therefore they must be considered as one (together with their whole-life aspects) to reveal a true and comprehensive cost picture.

3.4 Incremental costs

Incremental costs are those costs that would be avoided if the service were not provided [8]. Thus, overheads, which are those costs that do not alter with changes to the investment proposal (usually associated with costs incurred centrally), are ignored in determining the incremental costs. Incremental costs are useful in determining the minimum costs used for tariffing. Pilgrim [3] suggests that there is a danger that a proposal can be inadvertently rejected due to the inclusion of overheads which overstate the costs.

3.5 Cost modelling during the life cycle

Some of the techniques and methods described above can involve complex analysis in order to ensure the desired levels of accuracy. However, Bell et al [4] suggest that, depending upon which phase of the life cycle is being considered, a number of assumptions may be applied that significantly reduces the problem of costing. Concentrating upon the planning rule and network design phase for example, a number of assumptions may be invoked. In this case, since the development of the technology has already occurred, and interest is specifically in the planning rules, these costs may be neglected. Further, since again the interest is specific to the planning rules for the deployment of a particular technology, a number of other costs, e.g. storage of equipment, some transport costs on deployment, some planning time, can be neglected. Bell et al [4] point out that in this way the problem may be simplified to a manageable level and tailored to a specific requirement.

It should be noted that developing these assumptions and reducing the cost problem at hand, in the manner suggested by Bell et al, does not imply ignoring some aspects of the costs totally. Rather, it is the process of bounding the particular problem that is to be considered. Indeed, all the costs should be included in any business case, and subsequent cost analysis may just consider a subset of these costs.

4. Cost analysis

Having discussed the crucial role played by cost studies during all the phases of an access network life cycle, together with some of the main principles and methodologies that may be used, this section will briefly highlight some of the cost analysis that has been performed within BT recently.

4.1 Technology choice during the specification phase

The perceived demand for broadband services such as video-on-demand, together with the exponential growth in the Internet and the associated craving for faster transfer rates, has brought the discussion of the optimum access architecture to the fore. Quayle et al [15] describe most of the different architectures that are appropriate to an incumbent operator such as BT, to deliver these broadband services. Figure 2 illustrates typical IFC trends that will be borne by an incumbent PTT with a large copper base, that is considering deploying these different broadband architectures.

As is observed in Fig 2, the costs of all the different access network options decrease with increasing penetration (service/network take-up). This is, in the main, due to better utilisation of the equipment as more connections are required. However, perhaps more crucial is the fact that the fibre to the exchange (FTTExch) architecture using DSL technologies is, in general, relatively cheaper than the fibre to the building (FTTB), and the fibre to the kerb/cabinet (FTTK/Cab) options. The main reason for this is that the relatively expensive deployment of fibre in the access network is avoided and use is made of the large existing copper base to deliver the broadband service. Furthermore, the FTTK/Cab architectures tend to be less expensive from an IFC perspective than the FTTB options due to the reuse of the existing copper [16—19].

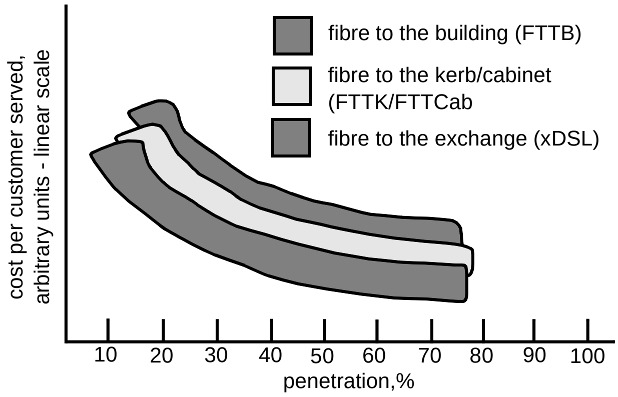


Fig 2 The variation of the installed first cost associated with a variety of broadband access architectures as the penetration increases.

However, it is worth noting that there can be significant overlap of the costs of these three options. Furthermore, a number of key factors may mean that FTTK/Cab architectures in particular are more expensive than FTTB. These factors include:

- equipment capacity and dimensioning,
- equipment and infrastructure cost,
- duct build requirements,
- labour costs.

Moreover, it is worth pointing out that while the FTTExch architecture utilising the latest DSL technology over the existing copper has an IFC advantage, the performance capabilities of all the different architectures must be considered before any holistic technology and architecture choices can be determined. For example, there are bandwidth limitations associated with the FTTExch and FTTK/Cab options that are not prevalent with FTTB architectures (and hence necessitates FTTB deployment for some customers). Nevertheless, there is significant interest in DSL technology with a significant number of network operators, including BT, adopting the technology in the access network [20—22].

4.2 Cost analysis during the network design phase

The necessity to deploy fibre in the access network requires significant cost modelling during the design phase. This is especially true when considering the physical infrastructure associated with passive optical networks (PONs) of the form discussed by Quayle et al [15]. The advent of TPON [19, 22] and OTIAN® technology [23, 24] has enabled BT to gain considerable experience of modelling the physical infrastructure costs of PONs in order to develop planning rules for minimum cost deployment.

Figure 3 illustrates the generic architecture associated with deploying PONs in the access network. As is observed, the network can be split into four locations — the exchange, the primary network, the secondary network and the customer drop network. The primary network is such that it may consist of either a ring or spine architecture, while two flexibility points are provided at the primary and the secondary node. A PON may consist of either a one- or two-level split with splitters located at either or both of the primary and secondary nodes.

When designing PONs of this nature for minimum cost, there are a number of factors to consider:

- the size and position of the primary split,
- the size and position of the secondary split,
- length and dimension of the primary, secondary and drop cables.

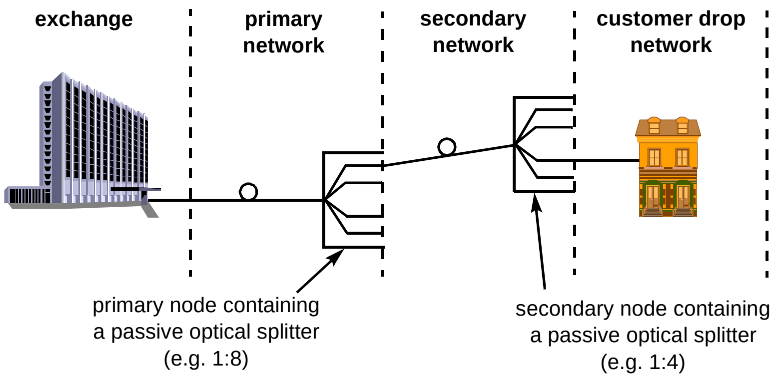


Fig 3 Generic PON architecture for the access network.

These factors generally have a conflicting relationship. For example, to minimise split cost it is imperative to achieve 100% port utilisation. However, because of the variance in the location and density of customer sites, it is not always possible to do this without consolidating the splitter at some distance from the customer base due to the tree-and-branch nature of the network. As a consequence, the secondary network and/or the customer drop length and cable dimension increases, thus adding to the cost of cabling. Conversely, if the cable costs are minimised this requires the splitters to be located close to customers but may result in poor port utilisation. Hence, an optimum balance needs to be sought between cost and number of splitters and the cost and size/length of cable. Figures 4 and 5 illustrate typical cost curves observed when these conflicting factors are considered, for a two-level split network.

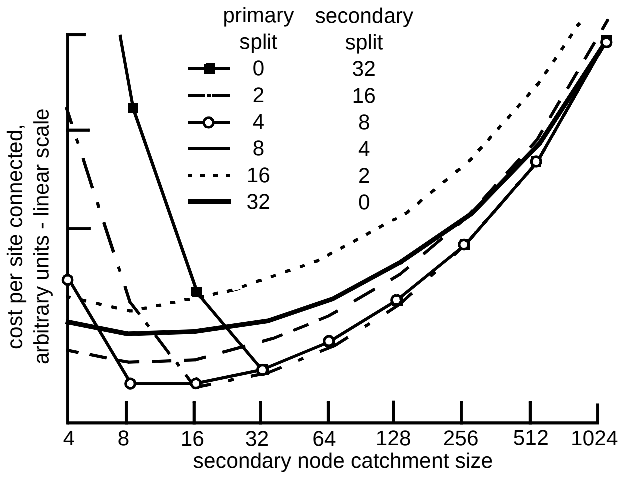


Fig 4 The deployment cost sensitivities of different split configurations and secondary node catchment sizes.

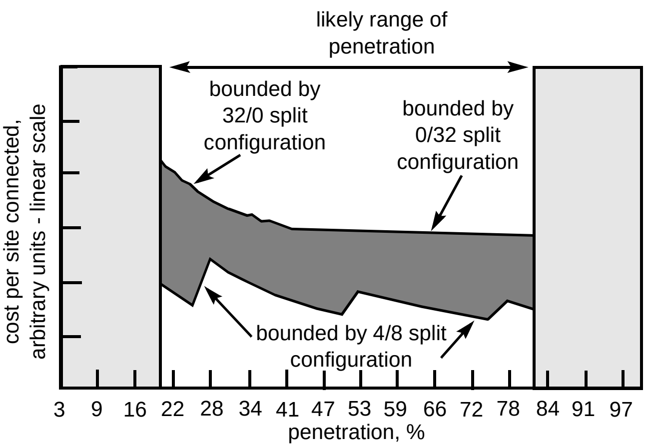


Fig 5 The effect of penetration (take-up) upon fibre infrastructure costs, where n_1/n_2 refers to a first level of split using a $1 \times n_1$ splitter, and a second level of split using a $1 \times n_2$ splitter.

Figure 4 illustrates both the effect that catchment size of the secondary splitter location has upon the IFCs, i.e. how the costs vary as the splitter is moved to capture more customer sites, and typical results from this modelling [4, 11, 12]. Here a maximum overall split of 32 is assumed, made up of a variety of split configurations. The catchment size is varied for these different configurations. As shown in Fig 4, the costs are relatively insensitive to a secondary node catchment size of between 8 to 32. Furthermore, the 4/8 and 2/16 split configurations are found to minimise these costs.

Figure 5 again details the costs associated with deploying 32-way PON; however, the impact of penetration (take-up of service) is considered and comparisons between the different 32-way PON configurations are highlighted. The use of a 2×4 splitter as the primary split and a 2×8 splitter as the secondary split provides an optimum cost solution over the range of 20% to 80% penetration.

It is analysis of this nature and similar analysis performed by Lea-Wilson [25] and Dalgoutte and Lea-Wilson [26] that aids the process of optimising the network costs during the design phase. However, as mentioned in section 2, the infrastructure deployment costs must not be considered in isolation. The whole-life and whole-network costs must be incorporated in any proposed planning and design rules.

Finally, it is worth pointing out that the increased interest in computer aided design (CAD) and computer aided modelling (CAM) has meant that a multitude of software tools are available to simplify and automate this modelling process [27, 28].

4.3 Process modelling during the in-service phase

The make-up of, and the sensitivities associated with, the service provision and repair processes of the access network have increased in proportion to the realisation of the crucial role that in-service or current account costs have upon the commercial viability of the access network [29].

BT has developed a model to investigate these process costs. Figure 6 shows the key parameters that have to be taken into account in building this dynamic model. These are the resources available to operate the processes, the definition of the processes themselves, and the network of interest. The key outputs from the model are the overall quality of service delivered to customers and the cost of the available resources.

Dynamic process simulation, employing queuing theory, is the method currently used within BT to implement a process model. The inputs to the model will be statistical distributions describing, for example, typical fault volumes, fault clearance times, travelling times, the number of new service requests and number of engineers available. The outputs from the model will be business performance measures such as the percentage of new service provisions that are within the time-scale agreed with the customer, the mean time taken to clear each fault and the percentage of faults cleared within a given target time.

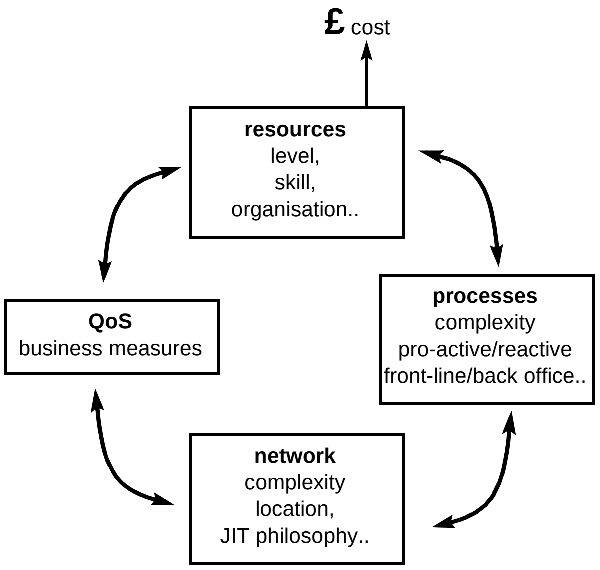
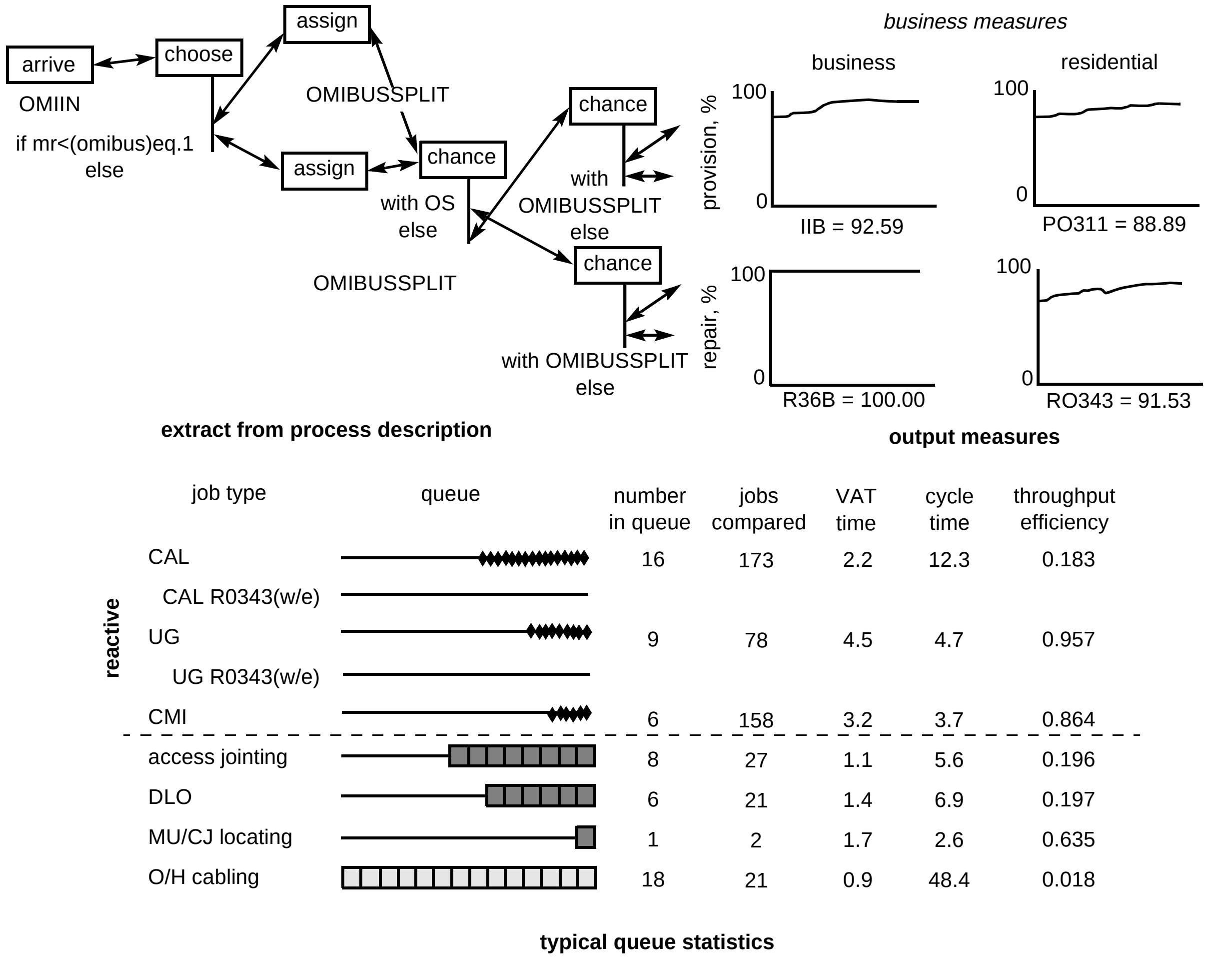


Fig 6 The key parameters in a process cost model.

Unlike the ABC methodology described earlier, the dynamic process simulation approach allows as yet un-implemented processes to be analysed. Another key advantage it offers over ABC is the ability to understand the relationships between cost and quality of service — for example, answering questions such as: ‘If we remove this process step, what difference will it make to quality of service?’ or: ‘If I reduce the number of engineers available by 10%, does customer service fall below contractual obligations?’

Figure 7 shows an extract from a typical simulation. The main features to note are the network of process steps that make up the model, graphical representation of key quality of service parameters and queuing statistics that assist in understanding the internal behaviour of the process.

While results of this nature provide invaluable information in understanding the quality of service associated with a given level of resource, it is often of interest to find the reverse relationship — how much resource is needed for a given quality of service? This ‘inverse resource function’ is far more difficult to determine using existing simulation techniques, with manually intensive iterative techniques usually being required. A



job type	number in queue	jobs compared	VAT time	cycle time	throughput efficiency
CAL	16	173	2.2	12.3	0.183
CAL R0343(w/e)					
UG	9	78	4.5	4.7	0.957
UG R0343(w/e)					
CMI	6	158	3.2	3.7	0.864
access jointing	8	27	1.1	5.6	0.196
DLO	6	21	1.4	6.9	0.197
MU/CJ locating	1	2	1.7	2.6	0.635
O/H cabling	18	21	0.9	48.4	0.018

Fig 7 Key features of the dynamic process model.

method of directly determining the resource requirement for a particular quality of service is the next mathematical challenge facing process modelling.

An additional enhancement of the process modelling approach is to associate process performance (quality of service) figures with customer satisfaction measures. This allows the impact of process changes on customer satisfaction — arguably the most important measure of all — to be estimated. Work on quantifying the relationship between quality of service and customer satisfaction is currently under way within BT.

5. Conclusions

This paper has considered the impact that costs have upon any proposed access network technology. Using the life cycle of any telecommunications network, the paper has indicated the importance of analysing not just the installed first costs, but all those costs associated with the technology throughout its whole life. This includes the development costs during the requirements, specification and network design phases to the in-service and de-commissioning costs. It is only by adopting this holistic approach that an overall cost optimum will be achieved.

The paper has also indicated some of the tools and principles that should be used by a cost modeller in order to fully understand the access costs. These include whole-life, end-to-end and activity-based costings. While these methods and concepts may at first seem complex, a number of simplifications can be invoked in order to provide a manageable problem.

Finally, it is hoped that this paper has given a flavour of some of the complex questions associated with access network cost analysis. Only by fully investigating these complex questions, can the full potential of the latest access technologies be unleashed. It is this challenge above all else which keeps access cost modellers engaged.

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